Task 3 and 4

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Task 3 and 4

1. You have learned the followings:
   a) BlockChain/DLT, Smart Contract, dAPP, DAO
   b) Tokenization, Digital Assets, Crypto-currencies, FT and NFT
   c) eRMB/eHKD, CBDC, DCEP
   d) Crypto Exchanges (SFC regulation), DeFi Exchanges
   e) Open Banking and APIs, FPS
   f) Digital ID, SSI, ZKP
   g) AI, Algorithms, IoT, IoV
   h) LoNG PESTEL, BIDT
2. Design a Web3 FinTech ecosystem with DLT applications, i.e. Smart Banking and Innovative Finance (SBIF)/POC (Proof of Concept) solution.

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Task 3 and 4 – Case Example

Design a Web3 SBIF FinTech ecosystem, e.g.

a) SBIF platform for the buyers/investors/sponsors/donors /lenders to meet with the sellers/goods & services providers/beneficiaries/borrowers
b) Identify the goods and services providers involved
c) Smart contract to manage the money/token/digital asset flows

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Task 3 and 4 – Case Example (con't)

Use the BIDT model to explain your design with diagrams/charts and words.

Case example:

B - Business idea/model/problem

Purpose: Raise money to start a florist shop business with prepaid coupon token sales

- Florist shop is trying to raise $1500 by selling 100 prepaid coupon tokens
- Each prepaid coupon token can redeem a dozen of rose in 1 month and will expire after 12 months
- Each prepaid coupon is selling for $15

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Task 3 and 4 – Case Example (con't)

D – Data technology/management/anlaytics

- Accepts eHKD, eCNY, HKD stablecoins, USDC, Bitcoin, etc (fungible tokens/FT) via credit cards, FPS, CeFi or DeFi Ex
- Creates 100 coupon tokens (FT)

T – Technology

- Use various AI tools to assist the design and development (https://builder.steps.media/)
- Use/Create DLT/BlockChain (Type III public, permissioned) to record all token transactions
- Use RegTech tools (i.e. https://www.chainalysis.com/) or Open Banking API services for AML check of the buyer's public key

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Task 3 and 4 – Case Example (con't)

T – Technology (con't)

Generate Smart Contract

- Buyer/Sponsor purchases 1 coupon token for $15 or equivalent using Open Banking API/TSP digital wallet or FPS (BaaS)
- When the token coupon holder wants to redeem the token for flowers, $5 will be released to the florist shop's supplier, $2 will be released to the courier, etc
- After the flowers are delivered (use oracle - triggered by the courier), the remaining $8 will be released to the florist shop after deducting platform fee $0.45 (3% of $15)
- After 12 months (expiry date), $15 will be released to the florist shop after deducting platform fee $0.45 (3% of $15)

***In here, you can offer other items, services, etc for redemption at expiry.**

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Task 3 and 4 – Case Example (con't)

I – Innovative Integration

- Please see the high-level flow diagram(s) attached involving the fintech/SBIF platform providing DeFi services to the clients (florist shop/merchant, sponsors/loyal customers) with DLT/BlockChain, Smart Contract and APIs to CeFi services (Open banking APIs/BaaS)

***In your design, you can add the coupon token buyer requests for business registration proof, rental receipt, business insurance certificates, etc. using SSI and ZKP technology.**

If so, tell me the additional components in the revised design…..

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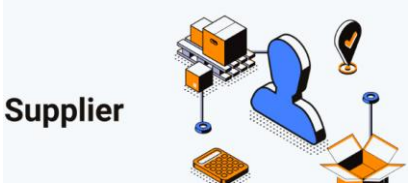
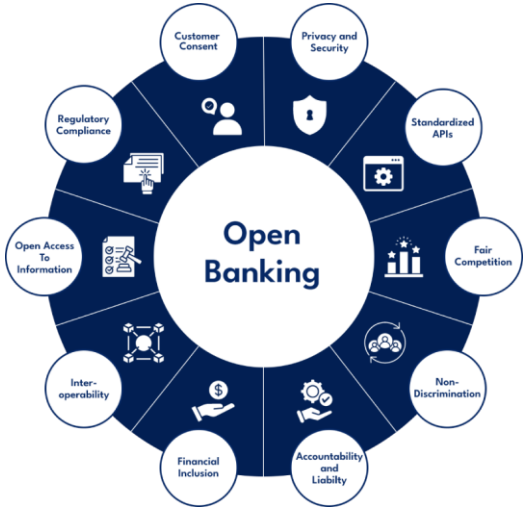
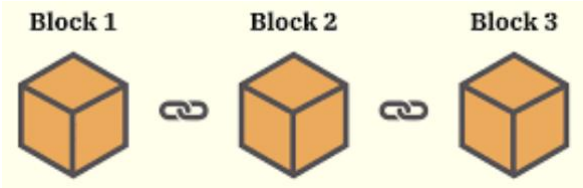
Use ER and flow diagrams to explain the entity-relationships, steps and flows







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ER and Flow Diagrams

Consider how you use the ER and Flow Diagrams to explain the case:

1. Creation of the prepaid coupon token
2. Purchase of the prepaid coupon token
3. Redemption of the prepaid coupon token
4. Smart Contract
5. …..

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DeFi vs DeFi

1. Go to https://builder.steps.media/ (STEPS) and use the template or AI to build your Web 3 services (landing page(s)) to support the POC of your design:
   1. Web3 fintech, mobile banking, p2p finance, etc.
   2. Web3 florist shop, cake shop, etc.
   3. Web3 AirBnb, travel agent, Uber/Grab, online games, etc.

   ***You just need to build a landing page. Of course you can do more to earn bonus marks**
2. Go to https://thirdweb.com/ and issue your own tokens for your Web 3 service.

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Team Submission

Describe and explain your Web3 Smart Banking Proposal/FinTech ecosystem in 2-3 pages (~500 words), excluding diagrams, charts and screen captures.

- Use the BIDT framework (refer to previous slides)
- What is the purpose and objectives of your SBIF proposal?
- What are the smart banking and innovative financial services that you propose to offer?
- Explain the ER and Flow Diagrams
- Give the URLs of the Web3 sites
- Give the name and smart contract of your token
- Explain how did you list the exchange of your token with SSU in Uniswap
- In conclusion: compare it with the existing banking and finance landscape, any limitations, opportunites and risks

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